

## SAFETY DATA SHEET

Creation Date 05-March-2010

Revision Date 25-December-2021

Revision Number 5

### 1. Identification

**Product Name** Bis(2-ethylhexyl) adipate

**Cat No. :** AC402460000; AC402460010; AC402460025; AC402460050; AC402462500

**CAS-No** 103-23-1  
**Synonyms** Bis(2-ethylhexyl)hexanedioate

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

**Specific target organ toxicity - (repeated exposure)** Category 2  
Target Organs - Kidney, Liver.

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray

**Response**

Get medical advice/attention if you feel unwell

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                 | CAS-No   | Weight % |
|---------------------------|----------|----------|
| Bis(2-ethylhexyl) adipate | 103-23-1 | >95      |

### 4. First-aid measures

|                                        |                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.          |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.  |
| <b>Inhalation</b>                      | Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Get medical attention.                                                                           |
| <b>Most important symptoms/effects</b> | No information available.                                                                                                |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                    |

### 5. Fire-fighting measures

|                                         |                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                 |
| <b>Flash Point</b>                      | 196 °C / 384.8 °F                                                        |
| <b>Method -</b>                         | No information available                                                 |
| <b>Autoignition Temperature</b>         | 350 °C / 662 °F                                                          |
| <b>Explosion Limits</b>                 |                                                                          |
| <b>Upper</b>                            | No data available                                                        |
| <b>Lower</b>                            | No data available                                                        |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                 |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                 |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

Health  
2 \*

Flammability  
1

Instability  
0

Physical hazards  
N/A

## 6. Accidental release measures

### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required.

### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

### Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Storage.

Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents.

## 8. Exposure controls / personal protection

### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.

### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

#### Hand Protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Butyl rubber   | recommendations   |                 |                        |
| Nitrile rubber |                   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

#### **Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### **9. Physical and chemical properties**

|                                               |                                      |
|-----------------------------------------------|--------------------------------------|
| <b>Physical State</b>                         | Liquid                               |
| <b>Appearance</b>                             | Colorless                            |
| <b>Odor</b>                                   | Slight                               |
| <b>Odor Threshold</b>                         | No information available             |
| <b>pH</b>                                     | Not applicable                       |
| <b>Melting Point/Range</b>                    | -76 °C / -105 °F                     |
| <b>Boiling Point/Range</b>                    | 210 - 218 °C / 410 - 424 °F @ 5 mmHg |
| <b>Flash Point</b>                            | 196 °C / 384.8 °F                    |
| <b>Evaporation Rate</b>                       | No information available             |
| <b>Flammability (solid,gas)</b>               | Not applicable                       |
| <b>Flammability or explosive limits</b>       |                                      |
| Upper                                         | No data available                    |
| Lower                                         | No data available                    |
| <b>Vapor Pressure</b>                         | 0.021 hPa @ 100 °C                   |
| <b>Vapor Density</b>                          | 12.8                                 |
| <b>Specific Gravity</b>                       | 0.924                                |
| <b>Solubility</b>                             | Insoluble in water                   |
| <b>Partition coefficient; n-octanol/water</b> | No data available                    |
| <b>Autoignition Temperature</b>               | 350 °C / 662 °F                      |
| <b>Decomposition Temperature</b>              | No information available             |
| <b>Viscosity</b>                              | 13-15 mPa.s @ 20°C                   |
| <b>Molecular Formula</b>                      | C22 H42 O4                           |
| <b>Molecular Weight</b>                       | 370.57                               |

### **10. Stability and reactivity**

|                                         |                                                         |
|-----------------------------------------|---------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available              |
| <b>Stability</b>                        | Stable under normal conditions.                         |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat.                     |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                 |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ) |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                |
| <b>Hazardous Reactions</b>              | None under normal processing.                           |

### **11. Toxicological information**

#### **Acute Toxicity**

#### **Product Information**

**Component Information**

| Component                 | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation             |
|---------------------------|---------------------------|------------------------------|-----------------------------|
| Bis(2-ethylhexyl) adipate | LD50 = 5600 mg/kg ( Rat ) | LD50 = 8410 mg/kg ( Rabbit ) | LC50 > 5.7 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** .

| Component                 | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------|----------|------------|------------|------------|------------|------------|
| Bis(2-ethylhexyl) adipate | 103-23-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** Kidney Liver

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information**

| Component                 | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|---------------------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Bis(2-ethylhexyl) adipate | Group III Chemical                       | Not applicable                                   | Not applicable                          |

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

**Ecotoxicity**

Do not empty into drains.

| Component                 | Freshwater Algae                                | Freshwater Fish                                                                                                                                                                  | Microtox   | Water Flea                            |
|---------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------|
| Bis(2-ethylhexyl) adipate | EC50: > 500 mg/L, 72h (Desmodesmus subspicatus) | LC50: 0.48 - 0.85 mg/L, 96h static (Pimephales promelas)<br>LC50: 0.48 - 0.85 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: 0.48 - 0.85 mg/L, 96h static (Lepomis macrochirus) | Not listed | EC50: > 1.6 mg/L, 48h (Daphnia magna) |

**Persistence and Degradability** Insoluble in water May persist based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component                 | log Pow |
|---------------------------|---------|
| Bis(2-ethylhexyl) adipate | 8.114   |

### 13. Disposal considerations

#### Waste Disposal Methods

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

|                 |               |
|-----------------|---------------|
| <u>DOT</u>      | Not regulated |
| <u>TDG</u>      | Not regulated |
| <u>IATA</u>     | Not regulated |
| <u>IMDG/IMO</u> | Not regulated |

### 15. Regulatory information

#### International Inventories

| Component                 | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|---------------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Bis(2-ethylhexyl) adipate | 103-23-1 | X   | -    | X    | ACTIVE                                        | 203-090-1 | -      | -   |

| Component                 | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------------|----------|-------|----------|------|------|------|------|-------|-------|
| Bis(2-ethylhexyl) adipate | 103-23-1 | X     | KE-18680 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component                 | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA)         |
|---------------------------|------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------|
| Bis(2-ethylhexyl) adipate | Part 1, Group A Substance                            | Schedule I                                                                 | Subject to Monitoring and Surveillance Activities |

#### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                 | CAS-No   | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------|----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Bis(2-ethylhexyl) adipate | 103-23-1 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No | Seveso III Directive (2012/18/EC) - | Seveso III Directive (2012/18/EC) - | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|--------|-------------------------------------|-------------------------------------|----------------------------|------------------------------------|
|-----------|--------|-------------------------------------|-------------------------------------|----------------------------|------------------------------------|

|                           |          | Qualifying Quantities<br>for Major Accident<br>Notification | Qualifying Quantities<br>for Safety Report<br>Requirements |                |                |
|---------------------------|----------|-------------------------------------------------------------|------------------------------------------------------------|----------------|----------------|
| Bis(2-ethylhexyl) adipate | 103-23-1 | Not applicable                                              | Not applicable                                             | Not applicable | Not applicable |

## 16. Other information

|                  |                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prepared By      | Regulatory Affairs<br>Thermo Fisher Scientific<br>Email: EMSDS.RA@thermofisher.com                                                                                                 |
| Creation Date    | 05-March-2010                                                                                                                                                                      |
| Revision Date    | 25-December-2021                                                                                                                                                                   |
| Print Date       | 25-December-2021                                                                                                                                                                   |
| Revision Summary | This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals. |

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**